

ALY 6110: Data Management & Big Data

Module 4 Final Project

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Boston 311 Service Request Analysis

Executive Summary

This analysis examines 246,271 Boston 311 service requests from 2025 to answer: "**How can Boston optimize its 311 service request system to improve response times and resource allocation across different neighborhoods and seasons?**" Using Databricks/PySpark and Power BI, three critical insights emerged: geographic clustering reveals Dorchester (31,010 requests) and Roxbury (22,521 requests, including 3,238 needle pickups) require targeted resources; department performance shows dramatic disparities with BTDT at 32.7% on-time versus PWDx at 88.1%; and temporal patterns demonstrate 48.8% volume increase from January to September with 85.83% digital adoption.

Methodology

The dataset from Boston's Open Data Portal contains 246,271 records with 30 variables including temporal data, geographic identifiers, request details, and submission sources. Databricks processed the data, creating calculated fields (resolution_days, month, day_of_week) and performing aggregations by neighborhood, department, and time period. Power BI visualizations include KPI cards, trend lines, bar charts, scatter plots, and interactive slicers enabling dynamic exploration.

Insight 1: Geographic Clustering

Geographic analysis reveals concentrated demand in specific neighborhoods with unique problem profiles. Dorchester leads with 31,010 requests (12.6% of total), followed by Roxbury

with 22,521 requests featuring a critical concentration of 3,238 needle pickups, the highest citywide. South Boston ranks third with 19,890 requests dominated by parking enforcement. The top 10 neighborhoods account for 69.8% of total volume.

The dashboard's scatter plot "Neighborhood Volume vs Response Time" displays bubbles sized by volume positioned by resolution speed, showing that high volume doesn't necessarily predict slow response. Dorchester's large bubble shows moderate 4.7-day resolution, while smaller neighborhoods demonstrate faster service. The neighborhood slicer enables filtering to reveal specific area patterns, such as Roxbury's disproportionate needle pickup crisis.

Recommendations include deploying specialized cleanup crews to Roxbury for public health response, establishing neighborhood coordinators for top three high-volume areas, and using quarterly scatter plot analysis to monitor service equity across all communities.

Insight 2: Performance Disparities

Department performance analysis reveals that organizational efficiency, not volume, determines speed. Citywide metrics show 3.99-day average resolution with 69% on-time completion. However, PWDx handles 118,172 requests (48% of total) with 1.7-day average and 88.1% on-time rate, while BTDT averages 9.9 days with only 32.7% on-time on 28,195 requests. ONS_ takes 36 days average with 20.4% on-time completion.

Request type analysis shows infrastructure issues take longest: plumbing averages 56.5 days, sign repairs 40.8 days (3,287 requests), and electrical work exceeds 35 days. The "Slowest Departments" bar chart immediately identifies bottlenecks, with variation from 1.7 to 36 days highlighting systemic inconsistencies.

Recommendations include immediate BTDT process audit to address 32.7% on-time performance, replicating PWDx's efficient practices across departments, and creating fast-track teams for infrastructure requests exceeding 35 days.

Insight 3: Temporal Patterns

Seasonal analysis reveals predictable fluctuations: January shows lowest volume at 18,603 requests while September peaks at 27,704 - a 48.8% increase. Weekly patterns show Tuesday peaks at 41,351 requests, with weekdays (Monday-Wednesday) accounting for 49.5% of weekly volume. Weekend volumes drop 35-40% to approximately 24,000-26,000 requests.

Digital transformation succeeds with 85.83% digital submissions: Employee Generated (46.8%), Citizens Connect App (29.1%), and City Worker App (9.9%). Traditional phone calls represent only 12.8%. The stacked area chart shows Employee Generated and App channels growing throughout 2025 while phone calls remain flat.

The monthly trend line clearly visualizes seasonal patterns enabling planning, while the submission source chart validates technology investments. The month slicer enables comparative analysis between seasons.

Recommendations include implementing dynamic staffing (40% increase June-September, 20% reduction December-February), concentrating resources Tuesday-Wednesday for demand peaks, pre-positioning assets in August for September peak, and continuing mobile app enhancements while gradually reducing phone support infrastructure.

Dashboard Design

The interactive dashboard contains six visualizations and three slicers. Four KPI cards provide instant metrics: Total Requests (246K), Average Resolution Days (3.99), On-Time Rate (69%), and Digital Rate (85.83%). The monthly trend line shows seasonal patterns, while the top 10 request types bar chart identifies common issues. The slowest departments chart highlights performance gaps, and the scatter plot reveals neighborhood volume-speed relationships. The submission source stacked area tracks digital adoption.

Interactive slicers for neighborhood, department, and month enable dynamic filtering with full cross-filtering functionality. Clicking any element filters the entire dashboard, supporting multidimensional analysis.

Recommendations Summary

Geographic optimization requires specialized teams for Roxbury's needle pickup crisis, neighborhood coordinators for high-volume areas, and quarterly equity monitoring. Performance improvement demands BTDT process overhaul, PWDx best practice replication, and infrastructure fast-track teams. Temporal management needs seasonal staffing adjustments (40% summer increase), weekday focus on Tuesday-Wednesday peaks, and August resource prepositioning. Technology advancement should enhance mobile apps, expand employee portals, and phase down phone infrastructure while maintaining accessibility.

Conclusion

This analysis of 246,271 Boston service requests demonstrates big data analytics' power in municipal optimization. Through Databricks processing and Power BI visualization, three insights emerged: geographic clustering reveals targeted resource needs, performance analysis identifies replicable best practices and critical inefficiencies, and temporal patterns enable proactive planning. The interactive dashboard transforms data into actionable intelligence for improved efficiency, equity, and citizen satisfaction. Implementing these recommendations will enhance service delivery, optimize resources, and ensure equitable access across all Boston neighborhoods.

References

City of Boston. (2025). *311 service requests*. Boston Open Data Portal. https://data.boston.gov/dataset/311-service-requests